

Variational Autoencoder (VAE): Probabilistic Latent Space

- Goal:

Learn a smooth, continuous latent space where we can sample new data points.

- Key Idea:

Instead of encoding an input into one point, VAE encodes it into a probability distribution.

- Encoder (Probabilistic)

For input x :

$$\mu, \log \sigma^2 = \text{Encoder}(x)$$

This defines:

$$q(z|x) = N(\mu, \sigma^2)$$

- Reparameterization Trick

Sampling is made differentiable:

$$z = \mu + \sigma \cdot \epsilon \quad \text{where } \epsilon \sim N(0, I)$$

- Decoder

Reconstruct input from latent variable:

$$\hat{x} = \text{Decoder}(z)$$

- Loss Function

Total loss has two parts:

1. Reconstruction Loss

Ensures output matches input:

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|^2$$

2. KL Divergence

Forces latent distribution to be normal:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2)$$

• Final Objective

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}$$

• Conclusion:

- Latent space becomes smooth & structured
- Sampling for $N(0, I)$ produces valid new data
- Enables generation, not just reconstruction.